Dash Beavers

Executive Summary

Which buildings and neighborhoods are most needing heat pumps? A common barrier towards widespread heat pump adoption is the lack of general awareness of this technology, resulting in a situation where only certain communities fully unlock the extreme heat pump benefits. However, through targeted marketing campaigns, the heat pump industry can access a whole new market in demand of their services.

Through this project, I have successfully identified the buildings most in need of heat pumps within New York City, a city of over 8 million people. New York City has thousands of commercial and multifamily homes, making it nearly impossible and incredibly time-intensive for a human to identify a single building that would most benefit from heat pump technologies. However, the system I have created not just identifies these buildings but also locates specific neighborhoods where heat pump marketing could have the largest impact.

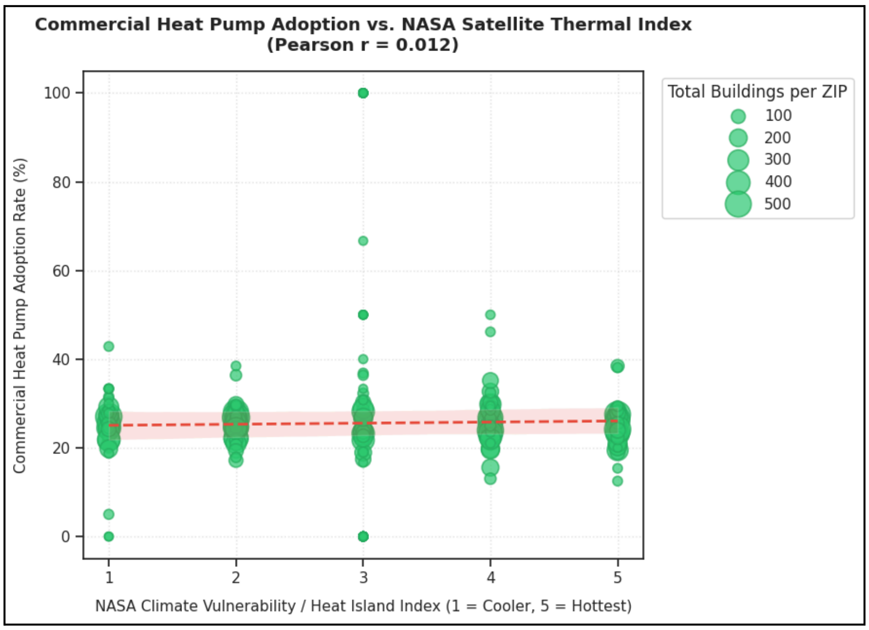
My server has been built to be used by those in the heat pump industry who are looking for potential clients that may currently be out-of-reach or unaware of the opportunities of heat pumps. By using my technology, heat pump professionals can easily reach out to building property managers with expiring HVAC systems in hotter areas of the city while also pinpointing specific areas of the city where marketing such as flyers or billboards may have the largest potential reward.

By easing the marketing of heat pumps, many benefits are realized. As heat pump marketing is more directed towards actual potential customers, the energy burden for these customers is significantly reduced, saving costs for people that may be left out of the building decarbonization transition. Moreover, by using electric heat pumps instead of natural gas furnaces, the neighborhood's disproportionate heating can be mitigated, resulting in downstream health benefits for those living and working in the area. Finally, by using heat pumps where people work and live instead of burning natural gas, toxins in the air can be lessened, reducing the likelihood of issues such as asthma or other detrimental respiratory diseases. Finally, by having a massive city transition to heat pumps, the city's carbon footprint significantly lowers, helping fight climate change.

Problem Statement

An issue currently plaguing urban decarbonization is that there is a severe disconnect between clean-energy technology providers and the specific buildings that urgently require their solutions. In New York City, and other dense metropolitan environments around the world, thousands of commercial and multifamily properties rely on aging, fossil-fuel-based HVAC systems. However, heat pump professionals and clean-tech firms lack the tools needed to efficiently and precisely locate these high-priority candidates. This gap in information results in a situation where those in the industry rely on an inefficient approach to marketing. This approach often results in low conversion rates, wastes deployment resources, and leaves communities entirely unaware of life-saving infrastructure upgrades.

When analyzing the live municipal building records across New York City's real estate ecosystem, the severe misallocation of clean infrastructure capital becomes mathematically clear. As shown in the statistical regression analysis below, done using the databases considered in the making of the system, there is a near-zero correlation ($r = 0.012$) between current commercial heat pump adoption rates and the NASA Climate Vulnerability Index. This flat line exposes a critical systemic failure: decarbonization efforts are completely stagnant and untargeted. Rather than capital flowing aggressively into the hottest, most vulnerable urban heat islands (Tiers 4 and 5) where cooling demands and utility costs are scaling unsustainably, heat pump deployments are scattered randomly across the city at a uniform, low baseline. Clean-tech solutions are failing to scale where they are needed most because the market lacks a coordinated router. Without a predictive data pipeline to guide operators, marketing campaigns remain blind to local climate stress, leaving the city's most critical high-heat corridors completely unaddressed.



It is crucial to time these infrastructure upgrades correctly if New York City wishes to reduce its emissions. By marketing to buildings that have recently installed a traditional combustion system, marketing resources will largely be wasted, as cost-aware buildings are not compelled to replace a new system. However, central mechanical heating and ventilation systems typically operate for 15- to 20- years before needing extensive repair or replacement. When legacy equipment fails, property managers frequently default to installing a familiar fossil-fuel replacement to quickly resolve the emergency, effectively locking the building into decades of continued carbon emissions and increased operational expenses. Furthermore, the continued reliance on traditional combustion systems in densely populated zones directly exacerbates the Urban Heat Island (UHI) effect. This escalates localized thermal stress, degrades outdoor air quality, and disproportionately burdens specific neighborhoods with higher cooling overheads and heightened respiratory health hazards.

This challenge persists because it is often incredibly difficult to identify project managers who may soon be searching for a new HVAC system before they properly enter the market following HVAC failure. The data required is vast, fragmented, and notoriously difficult for human teams to parse. Identifying an ideal, high-impact heat pump candidate requires cross-referencing municipal energy benchmarking registries, building age profiles, electrification proxies, and localized climate vulnerability metrics, such as NASA satellite thermal data. Historically, synthesizing these multi-layered structural and geospatial datasets into actionable, block-by-block sales leads has been too technically complex and time-intensive for standard marketing teams. Without AI-driven synthesis, the sheer scale of a city with over 8 million residents and thousands of commercial and multifamily buildings makes targeted intervention practically impossible.

Solving this market-targeting failure is fundamental to the future of the built environment. As municipalities push for stringent emission mandates, the inability to rapidly and equitably deploy clean heating and cooling technologies will stall broad decarbonization efforts. By bridging the gap between geospatial data and commercial outreach, we can vastly improve urban infrastructure resilience and environmental health. Equipping the heat pump industry with predictive, data-driven targeting ensures that capital is deployed exactly where it will maximize carbon reductions, alleviate grid strain, and eliminate toxic air pollutants in the communities that need it most.

Target Market

The direct, daily users interacting with this web application interface are clean-tech account executives and sales engineers who face massive informational asymmetry when hunting for commercial leads. Their core pain point is the extensive time and capital wasted cold-calling random property portfolios or blanketing broad neighborhoods with generic marketing without visibility into a building's real engineering lifecycle. By using this application, their workflow shifts from blind prospecting to executing a data-driven pipeline. It routes sales teams directly to specific buildings hitting their exact 15-to-20-year legacy HVAC failure windows, dramatically lowering client acquisition costs and accelerating clean-energy deployment velocities.

While property operations managers do not log into the software tool itself, they are the vital customer segment and ultimate decision-makers who must buy and adopt the heat pump technology. These stakeholders are traditionally risk-averse, financially anxious of sudden, mid-season mechanical breakdowns and mounting municipal emissions penalties. Their primary barrier to adoption is a general lack of trust or awareness of all-electric heating alternatives. The platform solves this friction point indirectly by empowering sales reps to deliver hyper-localized geospatial social proof. Showing an operations manager a similar building profile mere blocks away that has successfully electrified immediately lowers skepticism of a new technology.

The final, crucial stakeholders are the building occupants, urban residents, and utility grid operators who experience the direct downstream impacts of targeted electrification. By optimizing the deployment of heat pumps specifically within high-priority climate zones, the platform accelerates infrastructure upgrades where they will most effectively reduce ambient air toxins, energy insecurity, global warming, and high operational costs.

Solution Overview

The proposed solution is a live, automated software-as-a-service platform built specifically to streamline and scale the commercial building decarbonization pipeline. The core application functions as a predictive network that eliminates the traditional, time-intensive discovery barriers currently stalling the adoption of clean heating and cooling technologies. Rather than relying on static real estate logs or manual engineering audits, the system operates by continually utilizing data connected directly to live municipal databases. The interface presents a dual-page command deck designed to give clean-tech field representatives an instantaneous, targeted market map.

Instead of attempting to intervene after a building's heating infrastructure suffers an emergency mid-season failure, a point at which risk-averse managers almost always default to installing a legacy fossil fuel replacement, the application is designed to intercept operations managers soon before a mechanical crisis may occur. The pipeline applies an automated lifecycle filtering rule that isolates properties whose HVAC systems are currently hitting their 15-to-20 year capital replacement window. By identifying these critical, unelectrified building profiles as they begin the search for a new HVAC system, the platform provides clean-tech providers with an optimal intervention window to position electric heat pump technologies.

The platform uses a combination of deterministic data engineering and algorithmic optimization to evaluate properties and generate text that can be sent by a heat pump marketer to a prospective building client. Because municipal records rarely contain explicit tracking fields indicating a building's internal HVAC mechanics, the system uses an automated proxy. It evaluates building energy consumption data, systematically flagging a property as a baseline fossil-fuel structure if it demonstrates an active reliance on natural gas or fuel oil alongside electricity, or classifying it as an electrified baseline if its heating envelope relies purely on electricity. Concurrently, the engine executes a geospatial coordinate sweep that cross-references building coordinates against raw NASA ECOSTRESS satellite thermal index points. This step allows the model to prioritize buildings trapped inside extreme urban heat islands, where operational cooling costs are scaling unsustainably. Finally, the server runs a localized engine that calculates the exact distance between the selected target and the nearest electrified building in the database to generate an automated cold email, as shown in Figure 1 of the appendix. This outreach is embedded with hyper-localized neighborhood landmarks and street-level verification data to instantly eliminate client skepticism. The goal is to make it easier for heat pump marketers to easily communicate with building managers, eliminating an often tedious step of writing repetitive emails.

System Workflow

The software of the decarbonization server relies on a structured, automated routing and text-generation pipeline. The process initiates the moment a user chooses a target real estate classification into the frontend Search Finder tab. This classification can be “hotel”, “office”, “retail store”, “supermarket/grocery store”, “commercial” for all the previous classifications together, and “multifamily housing.” Behind the user interface, the server establishes an active connection to live municipal endpoints via an API gateway, streaming raw building records and energy tracking indices into temporary data frame memories. Rather than navigating these immense, unstructured row matrices manually, the server passes the active arrays into a multi-tiered data cleaning engine. This block automatically handles null missing coordinate variables, standardizes neighborhood tracking strings, and applies an aggressive deduplication layer based on unique street addresses to strip away data redundancy and prevent self-referencing communication loops.

Once the core database is sanitized, the system applies parallel algorithmic filters. It isolates properties whose central heating and ventilation systems fall into the critical 15-to-20-year operational lifecycle window while simultaneously evaluating energy consumption lines to flag non-electrified structures. This localized building target is then joined with regional NASA thermal layers, sorting the final buildings leads by vulnerability stress, as shown in Figure 2 of the appendix. Finally, when an operator selects an individual property from the compiled pipeline table, a proximity sweep script executes coordinate-distance vector calculations to identify the closest successful heat pump adopter. This near-neighbor anchor string is piped directly into a generative copywriting template alongside local landmark mappings, outputting a highly tailored cold outreach proposal while refreshing the dedicated, street-level map view panel. This map view panel is shown in Figure 3 of the appendix.

Flowchart

I. Inputs
Step 1: User Query → User representative enters building search classification
Step 2: Use API Data → Server uses active API data for energy/location data
Step 3: Deduplication → Server eliminates all duplicate locations

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II. AI Processing Layer
Step 4: Energy Proxy Rule → Server uses utility data to determine building electrification status
Step 5: Lifecycle Filter → Server identifies buildings with 15-20 year age range
Step 6: Heating Integration → Uses zip code data to identify building NASA ECOSTRESS zone
Step 7: Priority Scoring → Ranks community heat pump priority using Equation 1 in Appendix

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III. Outputs & Actions
Step 8: Proximity Scan → Identifies closest electrified building to target building
Step 9: Landmark Mapping → Matches building ZIP code with landmark dictionary
Step 10: Email Render → Writes the personalized marketing cold email
Step 11: Map Render → Creates map identifying target building and electrified building

The ten-step pipeline detailed above simultaneously drives both primary user interfaces of the application. Steps 1 through 7 establish a unified, backend data-processing core that handles data ingestion, AI fuel proxies, and NASA climate mapping citywide. Once this data matrix is compiled, the output layer branches dynamically based on the user's active interface tab. On Tab 1, titled "Building Demand Search," Steps 8 through 10 execute a micro-targeted coordinate sweep to generate individual cold emails and street-level proximity maps. Concurrently, Tab 2, titled

"Neighborhood Demand Dashboard," taps into this exact same processed engine array, aggregating the properties by postal code to populate the macro-level territory leaderboard, ranked according to Priority Score, and depict the citywide pie chart rings by percentage of buildings within the zip code that are electrified, as shown in Figure 3 and 4 of the appendix respectively.

Implementation

The platform will initially deploy as a cloud-based free tool tailored for clean-tech account executives and localized heat pump distributors targeting dense urban areas, potentially beyond just New York City, with robust energy-compliance mandates. By inputting parameters to isolate properties hitting their 15-to-20-year legacy HVAC failure windows, clean-tech sales teams can bypass manual real estate prospecting and rapidly build high-conversion lead pipelines. Once product-market fit is validated through initial project acquisitions, the commercial model will scale into a tiered, subscription Software-as-a-Service architecture that integrates premium database layers and automated lead webhooks directly into corporate software.

A primary market barrier identified by the citywide dataset is an environmental equity dilemma: the vast majority of high-priority, extreme urban heat islands requiring urgent HVAC retrofits are clustered in low-income, under-resourced neighborhoods of color. To overcome this capital hurdle, the platform's outreach pipeline can pair directly with a Pay As You Save (PAYS) tariff-based financing mechanism, a mechanism currently being formally explored by New York legislators.[1] This reduces the upfront cost burden of heat pump installation in low-income areas.

The internal AI proxy conservatively filters out hybrid buildings to shield field reps from chasing false leads, though it introduces a margin of false negatives. Ultimately, the server establishes a major competitive advantage over legacy real estate databases by dynamically fusing structural parameters with real-time NASA ECOSTRESS satellite thermal index points, transforming static public records into a closed-loop outreach system that drives equitable urban decarbonization.

Conclusion

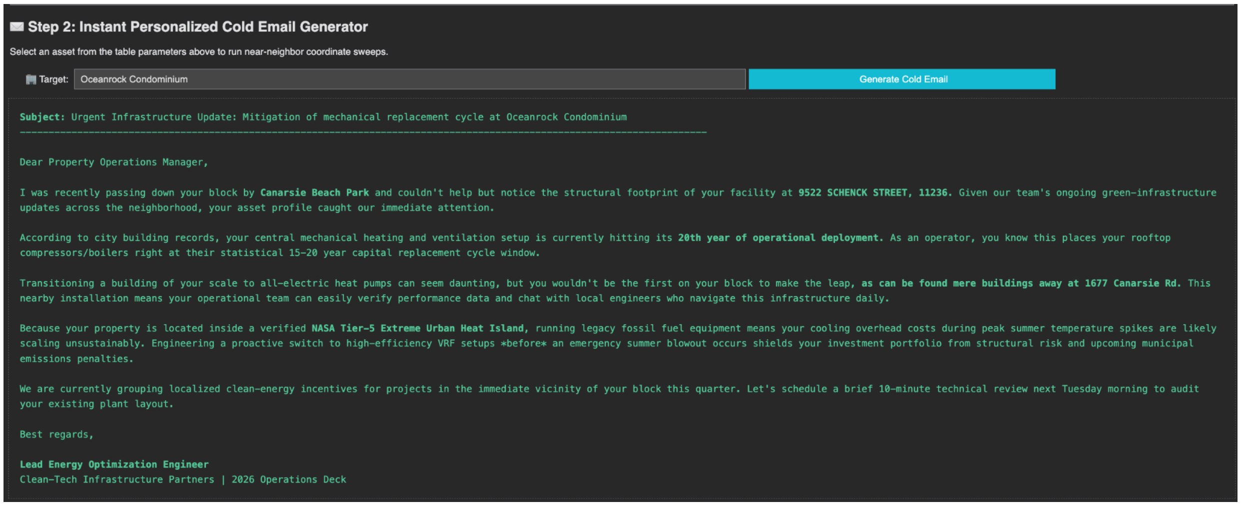
This predictive routing platform addresses an informational failure in market targeting and general public awareness. By transforming fragmented municipal compliance registries into a closed-loop system, the engine allows clean-tech providers to systematically intercept property managers exactly when an intervention maximizes capital efficiency and directly relieves environmental stress in vulnerable communities.

While optimized for New York City, the underlying architecture is highly modular and replicable across urban areas. Cities deploying progressive energy standards can seamlessly input their local real estate registries into the same algorithmic processing core. As long as a city maintains basic public records tracking building ages, coordinates, and utility fuel-mix indicators, this AI engine can be deployed citywide, serving as a globally scalable blueprint for accelerating equitable urban decarbonization.

[1]Clean Energy Works. "Introduction to Inclusive Utility Investments." *Clean Energy Works*, 1 Jan. 2023, www.cleanenergyworks.org/2023/01/01/introduction-to-inclusive-utility-investments/.

Appendix

Figure 1



Step 2: Instant Personalized Cold Email Generator

Select an asset from the table parameters above to run near-neighbor coordinate sweeps.

Target: Oceanrock Condominium | Generate Cold Email

Subject: Urgent Infrastructure Update: Mitigation of mechanical replacement cycle at Oceanrock Condominium

Dear Property Operations Manager,

I was recently passing down your block by **Canarsie Beach Park** and couldn't help but notice the structural footprint of your facility at **9522 SCHENCK STREET, 11236.** Given our team's ongoing green-infrastructure updates across the neighborhood, your asset profile caught our immediate attention.

According to city building records, your central mechanical heating and ventilation setup is currently hitting its **20th year of operational deployment.** As an operator, you know this places your rooftop compressors/boilers right at their statistical 15-20 year capital replacement cycle window.

Transitioning a building of your scale to all-electric heat pumps can seem daunting, but you wouldn't be the first on your block to make the leap, **as can be found mere buildings away at 1677 Canarsie Rd.** This nearby installation means your operational team can easily verify performance data and chat with local engineers who navigate this infrastructure daily.

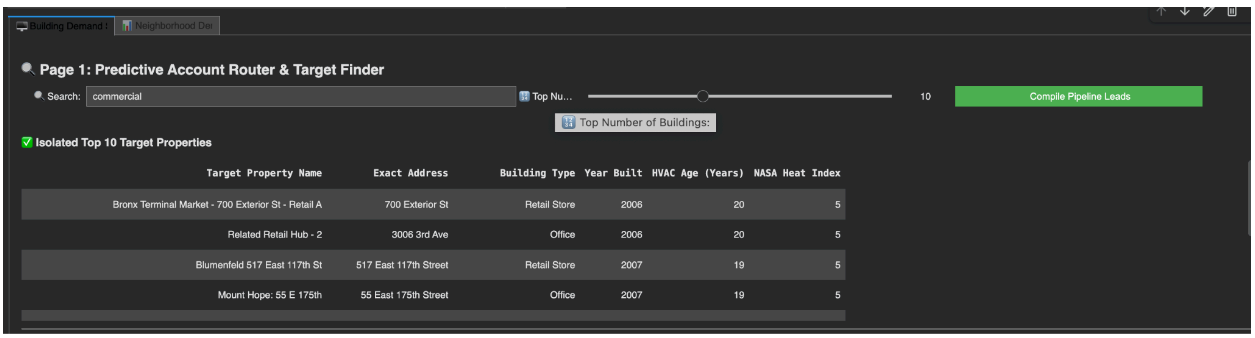
Because your property is located inside a verified **NASA Tier-5 Extreme Urban Heat Island,** running legacy fossil fuel equipment means your cooling overhead costs during peak summer temperature spikes are likely scaling unsustainably. Engineering a proactive switch to high-efficiency VRF setups *before* an emergency summer blowout occurs shields your investment portfolio from structural risk and upcoming municipal emissions penalties.

We are currently grouping localized clean-energy incentives for projects in the immediate vicinity of your block this quarter. Let's schedule a brief 10-minute technical review next Tuesday morning to audit your existing plant layout.

Best regards,

Lead Energy Optimization Engineer
Clean-Tech Infrastructure Partners | 2026 Operations Deck

Figure 2



Page 1: Predictive Account Router & Target Finder

Search: commercial | Top Nu... | 10 | Compile Pipeline Leads

Top Number of Buildings:

Isolated Top 10 Target Properties

Target Property Name	Exact Address	Building Type	Year Built	HVAC Age (Years)	NASA Heat Index
Bronx Terminal Market - 700 Exterior St - Retail A	700 Exterior St	Retail Store	2006	20	5
Related Retail Hub - 2	3006 3rd Ave	Office	2006	20	5
Blumenfeld 517 East 117th St	517 East 117th Street	Retail Store	2007	19	5
Mount Hope: 55 E 175th	55 East 175th Street	Office	2007	19	5

Figure 3

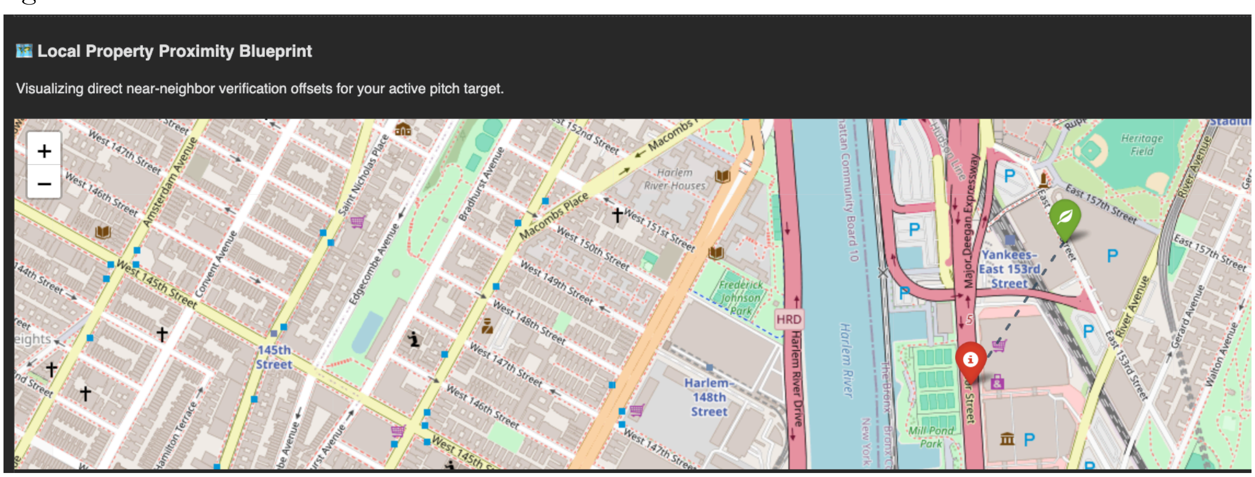
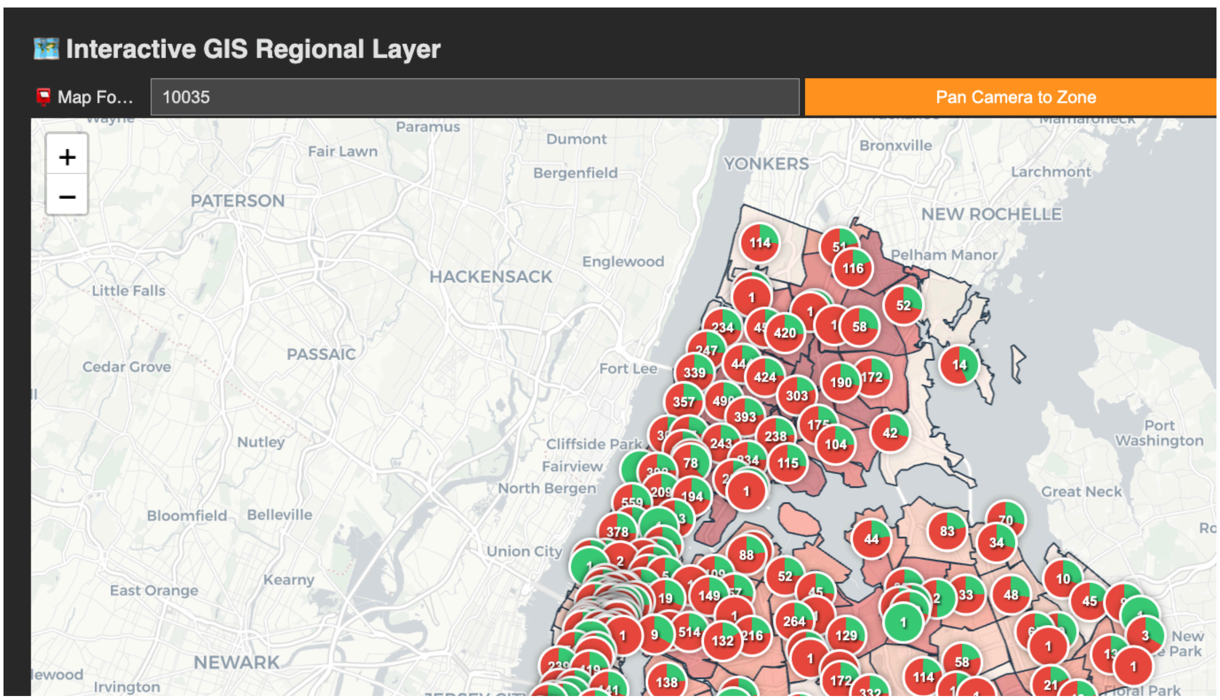


Figure 4

Citywide Leaderboard

ZIP Code	NASA Heat Index	Fossil Fuel Baselines	Priority Score
10451	5	27	135
11101	4	32	128
11211	3	36	108
10456	5	19	95
10459	5	19	95
11205	4	23	92
11354	4	23	92
11206	4	23	92
10457	5	18	90
10035	5	16	80
11249	3	26	78

Figure 5



Equation 1

Priority Score = A * B where:

A is NASA Heat Index Score from 1-5 (increasing numerically as heat increases)

B is Fossil Fuel Baselines (number of buildings within ZIP Code that have soon-to-expire gas HVAC systems)